

Sentiment Analysis on r/Drexel

What students discuss and how they feel

Group 7

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Our Research Question

What are students talking about on r/Drexel, and how do they feel about those topics?

- Reddit captures candid, real-time student discussion
- The data is too large and unstructured to read manually
- Topic modeling + sentiment analysis turn posts into usable patterns

Stakeholders

Who will benefit from this analysis?

- **Drexel Administration:**

Since students do not fill out official surveys, allows administration to identify issues and get unfiltered feedback.

- **Prospective Students:**

Make an objective decision by identifying good and bad things about the university.

- **Current Students:**

Get answers to questions and understand if they are the only ones struggling with something or if it is a shared experience.

The Dataset

 r/Drexel posts from subreddit creation through July 1, 2026

32,053

raw posts collected

22,132

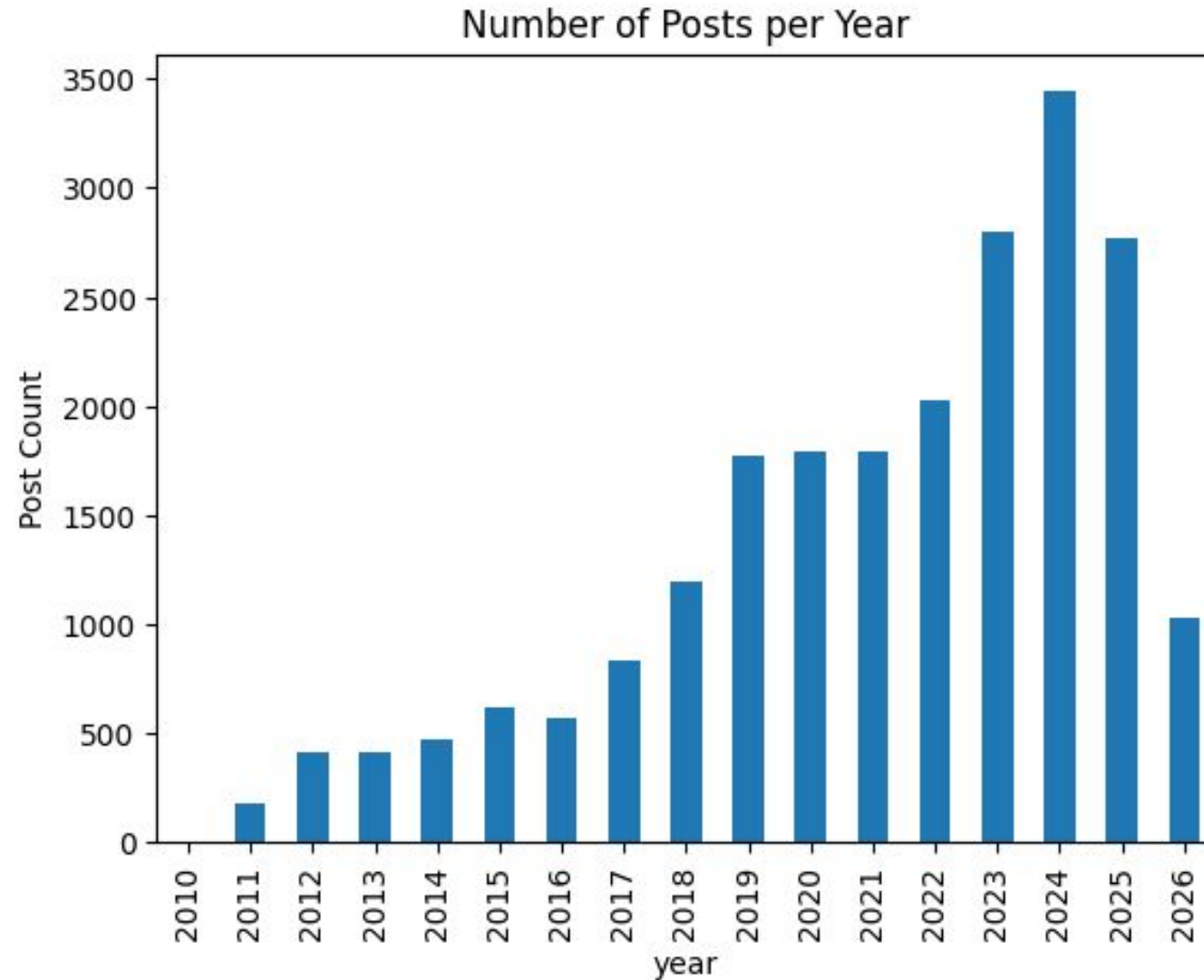
posts after filtering

8,983

unique authors

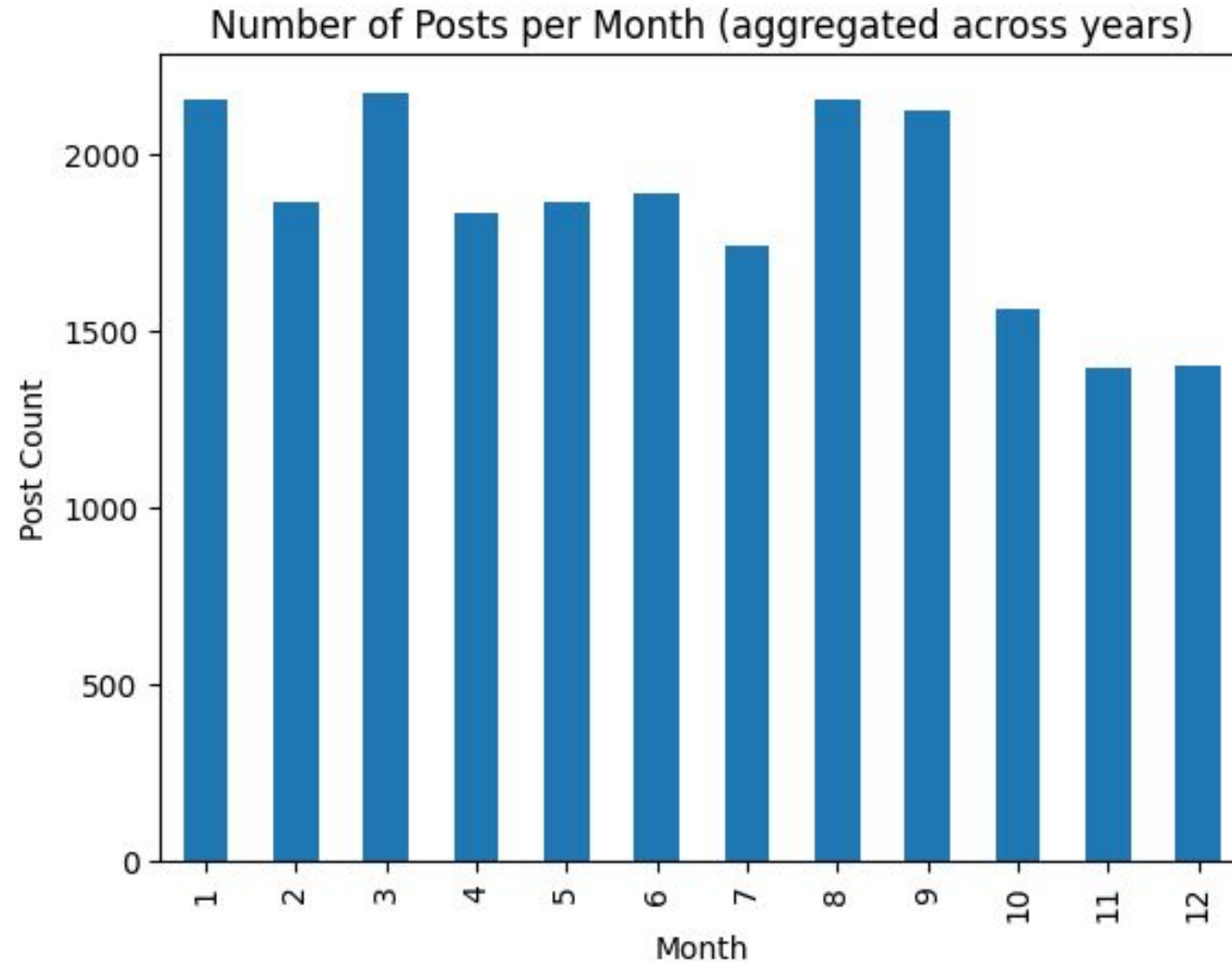
- Removed deleted/removed posts and kept id, author, date, title, and selftext
- 5,526 authors posted only once
- Top 1% of authors produced 12.45% of posts

Posting Activity Grew Over Time



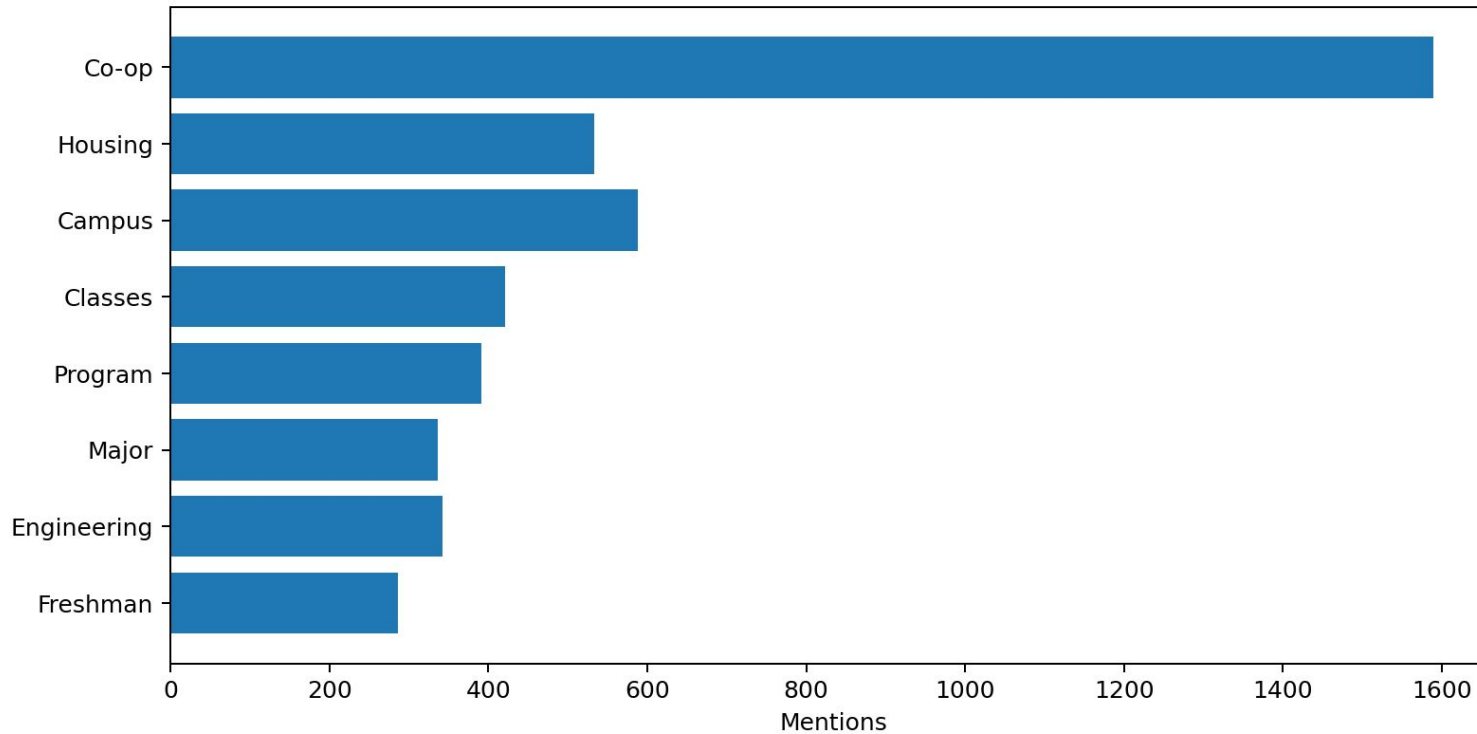
- Posting generally increased over the life of the subreddit
- 2024 was the highest-volume year in the notebook
- 2026 is only a partial year through July 1

When Do Students Post?



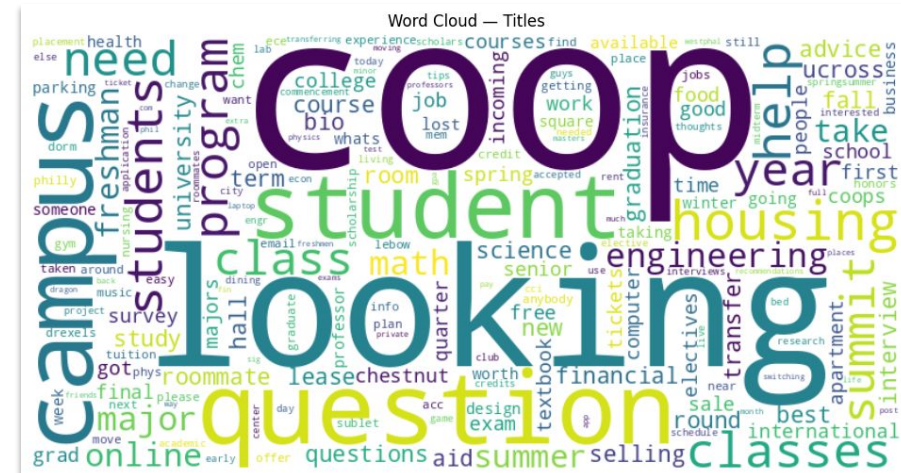
- Posting peaks appear in January, March, August, and September
- The group hypothesizes that some peaks may line up with co-op and academic decision periods
- Activity is also highest from roughly early afternoon through late evening

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most common meaningful word in both titles and post text

- Other repeated themes: classes, housing, campus, majors, programs, and engineering



Topic Modeling: 7 Recurring Themes

LDA with K = 7; final notebook coherence ≈ 0.554

1

Campus & community

campus • club • food

2

Co-op & jobs

interview • job • coop

3

Programs & aid

program • financial • transfer

4

Housing

apartment • lease • roommate

5

Academics

class • course • professor

6

Marketplace

sell • ticket • buy

7

Communication

email • send • issue

Sentiment Analysis with VADER



Positive

compound ≥ 0.05



Neutral

$-0.05 < \text{compound} < 0.05$



Negative

compound ≤ -0.05

What the code does

- Combines each post title + selftext
- Calculates negative, neutral, positive, and compound scores
- Assigns a sentiment label to every post

Why We Chose VADER

Naive baseline

Counted a short list of positive and negative words, then labeled ties as neutral

VADER model

Uses polarity strength, negation, punctuation, and common social-media wording

Why it matters

Reddit posts are informal, sarcastic, and often mixed in tone

53.8%

agreement with VADER

0.285

Cohen's kappa

12,458

posts naive called neutral

6,071

disagreements with negation

Main point: the naive model overused “neutral,” while VADER captured more real positive and negative signals.

Naive Word Counts Missed Sentiment

VADER vs. Naive Word-Count Labels

VADER negative	720	2,654	621
VADER neutral	31	3,232	50
VADER positive	301	6,572	7,951
	Naive negative	Naive neutral	Naive positive

9,226

VADER non-neutral posts that naive labeled neutral

- Naive missed many posts without exact dictionary words
- Negation changed meaning: “not,” “never,” “without,” and “no.”
- VADER was better for informal Reddit language

Examples

- Where do you guys find parking? I'm over near Powelton Avenue and I'm sick of getting parking tickets because I can never find a spot and end up in some temporary spot too long.\n\nWhere do you guys usually park? Or do you use Drexel Parking?\n\nThanks.

Naive Model: Positive, VADER: Negative (more accurate)

- Somehow some way Drexel makes housing even sh**** for upperclassman. my girlfriend just got an email from housing, shes gonna live in caneris, and she's gonna have a freshman stay in her suite in addition to her five other roommates, meaning the suite . for those counting at home she now has 7 people in her room, one of the rooms is a triple. she is kinda pi****.

Naive Model: Neutral, VADER: Negative (more accurate)

- The people who hog the Buckley soccer field every evening... I never see them anywhere else... in hallways, on campus, or in class...\n\nI'm not keeping an eye out for them directly but hopefully what I'm saying is understood.\n\nI see them arrive in cars, play, then leave. So they are either commuters (that I somehow don't see anywhere on campus) or not Drexel students (someone lets them in?)\n\nEither way it's kinda annoying when other people want to play soccer or any other sport (they always take up the whole field too)\n\n</rant>

Naive Model: Negative (more accurate), VADER: Positive

Next Steps and Future Research

- Compare sentiment across all 7 LDA topics to identify where student experiences are most positive or negative
- Validate a sample of posts by hand to measure VADER accuracy, especially for sarcasm, slang, and mixed tone
- Track topic and sentiment changes over time to connect shifts with academic, co-op, and housing cycles
- Use a stronger context-aware sentiment model as a comparison to VADER and report where the models disagree
- Translate recurring negative themes into specific areas Drexel can investigate through more targeted surveys or student outreach

Key Takeaways

- r/Drexel provides a large, candid source of student discussion that complements traditional feedback methods
- Co-op, academics, housing, and campus life are recurring concerns that can be organized into clear themes with LDA
- VADER captures more sentiment than the naive word-count approach, but sarcasm and informal language still create errors
- Combining Topic × Sentiment is the strongest next analysis because it shows not only what students discuss, but how they feel about each issue
- With validation and time-based analysis, these findings can better support administrators, current students, and prospective students

Final takeaway: Topic + sentiment can turn unstructured Reddit posts into useful signals for decision-making